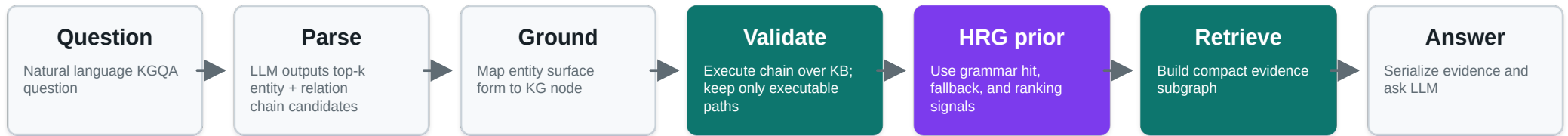


HRG-guided KG-RAG: offline grammar prior and online evidence retrieval

Goal: extract structural priors from the KG, then use them during QA to select compact, executable evidence.



What becomes explainable?

selected_entity

topic node grounded in KG

relation_labels

ordered KB relation labels

HRG signal

grammar hit / ranking prior

spine_edges

printable evidence triples

final_context

small context sent to LLM

Main evidence claim

MetaQA: BFS 4415.83 context tokens

HRG-Proposed-triple: 77.26 context tokens

HRG is evaluated as a structural prior, not only compression

Claim boundary

Not a claim of universal accuracy improvement

MLPQ / KQAPro remain hard cases

HRG is a structural prior for evidence retrieval